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From the black box to the glass box: Using unsupervised and supervised learning processes to predict user engagement for the airline companies

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ABSTRACT

Firms collect an enormous amount of user generated content (UGC), such as social media posts, to analyze consumers' unfiltered opinions regarding brands and firms. A challenge in analyzing unstructured UGC is the lack of analytic frame. By adopting both unsupervised and supervised learning processes for using artificial intelligence (AI), we collected 680,410, tweets related to airline companies (United Airlines, Delta Airlines, Southwest Airlines, Alaska Airlines, and Hawaiian Airlines) and analyzed 4961 retweets to predict user engagement levels on Twitter. Rooted in the electronic word-of-mouth (eWOM) perspective, the results of this study indicated that consumer sentiment was positive for United Airlines, Delta Airlines, and Alaska Airlines, whereas it was negative for Southwest Airlines and Hawaiian Airlines. We also examined the effects of word count, gaps between the tweet generated date and the retweeted date, the number of the hashtag(s), and extracted topics on predicting the level of user engagement. Ultimately, this study provided a detailed guide to mangers on how to use an unstructured data analysis procedure incorporating both supervised and unsupervised learning processes.

1. Introduction

The travel industry generated $1 trillion (10.3%) of the global gross domestic product (GDP) in 2021 and created 217 million jobs globally (Simpson, 2022). The airline industry alone supports 750,000 direct employees (Airlines for America, 2022), yet it has one of the poorest customer satisfaction rankings based on the American Customer Service Index (ACSI) (DiMeglio, 2022). For example, the ACSI rating for the airline industry was 75 in 2022, which was lower than the ratings for breweries (85), personal care and cleaning products (83), and food manufacturers (82), and 33% of business travelers and 11% of leisure travelers complained about their airline experience (American Customer Satisfaction Index, 2019). Customers' expenditures on airline tickets and the flight experience are key antecedents for consumers' satisfaction with their trips (Chu et al., 2017). Yet the consumer price index for airline tickets shot up 25 percent in 2022, the largest increase since the Federal Reserve of St. Louis began tracking the index in 1989 (Holzhauer, 2022). In addition, airlines have seen a dramatic increase in-flight incidents that have tainted the travel experience for all involved (Baldanza, 2021), and a rash of flight cancellations and delays have further sullied the reputations of airlines and tamped down customer satisfaction (Chen & Wichter, 2023; Brown & Cripps, 2022). Understandably, improving customer satisfaction is high on the agenda of U.S. airlines (NPS Benchmarks, 2023).

Conceptually, to address the airline industry's low customer satisfaction level, many studies have examined the antecedents of customer satisfaction by employing a variety of methodologies such as surveys using structured data (e.g., Babbar & Koufteros, 2008) or experiments (e.g., Hazée et al., 2017). However, such studies have largely neglected the recent advanced analytic framework of machine learning and big data analysis approaches from the information management literature (e.g., Ahn et al., 2021; Ge et al., 2022; Jeong et al., 2019; Pröllochs & Feuerriegel, 2020; Zhong, 2023). These approaches hold great promise for providing insights that will enable airlines to improve customer satisfaction dramatically.

Therefore, the objective of this research is to understand how consumers' opinions toward airlines are shaped across the different companies and how these opinions affect the level of user engagement on Twitter. Capturing consumers' unfiltered opinions is the key driving force behind successful marketing strategies. From developing new product

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Table 1
Relevant literature on airline industry management.

Author (Year)	Methodology	Context	Key outcomes	Data	Causal Inference
Ringle et al. (2011)	Survey	Factors influencing airline customers' satisfaction	Perceived safety is a positive predictor of customer satisfaction, and the purpose of travel type moderates this effect. For the trip for pleasure, the effect of perceived safety on customer satisfaction is amplified	Self-reported, 1031 participants	Yes
Misopoulos et al. (2014)	Social media data analysis from Twitter	Identifying elements of customer service in social media	Positive sentiments were linked mostly to online and mobile check-in services, favorable prices, and flight experiences, whereas negative sentiments were associated with the usability of companies' websites, flight delays, and lost luggage	67,953 publicly shared tweets	No
Palma-Mendoza et al. (2014)	Case study	Supply chain integration	Proposed methodology for Airline Maintenance Repair and Overhaul (MRO) supply chain	N.A	No
Homburg et al. (2015)	Social media data analysis from airline and hotel industry forum	Consumer sentiment	Consumers show diminishing returns to active firm engagement, which can undermine consumer sentiment	115,000 consumer posts in online forums	Yes
Xu et al. (2018)	Content analysis of customer review from Skytrax	Impacts of service failure and recovery efforts on airline customers' satisfaction and emotion	Providing compensation for the current trip can alleviate customers' negative emotions, whereas providing compensation for future travel does not have any impact on emotions	6500 online reviews	Yes
Dillette et al. (2018)	Qualitative	Tweets using the hashtag #TravelingWhileBlack	1) occurrences of racism, 2) awareness of being Black while traveling, and 3) meaningful experiences traveling while Black was the main theme of tweets	328 Tweets using a certain hashtag	No
Chen et al. (2019)	Survey	Mobile app adoption barrier among airline customers	Usage barrier, image barrier, and value varier are associated with resistance to mobile app adoption, and satisfaction with off-line service moderates this effect	Self-reported, 532 participants	Yes
Zhang et al. (2019)	Interview and survey	Consumers' trust in the voluntary carbon offsetting of the airline industry	Trustworthiness toward the airline industry's message source has a positive relationship with the purchase intention of aviation voluntary carbon offsetting (VCO) products	Self-reported, 1500 participants	Yes
Ioannou et al. (2020)	Survey	Travelers' privacy concern	Disposition to privacy, perceived privacy control, trust, privacy experience, and privacy protection regulation are predictors of personalization, travelers' privacy concern (TOPC)	Self-reported, 645 participants	Yes
Chi et al. (2020)	Survey	Consumers' perception toward the usage of artificially intelligence (AI) devices	Social influence, hedonic motivation, anthropomorphism, performance, effort expectancy, and emotion toward artificially intelligence devices are associated with the AI devices usage intention	Self-reported, 422 participants	Yes
Rahman (2020)	Secondary data	Dynamic marketing productivity (DMP) on airline companies' financial performance	DMP can be a source of competitive advantage that ultimately affects firms' financial performance	Single industry of the US airline industry	Yes
This study	Unsupervised learning and supervise learning	Identifying factors affecting customer engagement in Twitter for airline companies and estimating the effects of different topics discussed on Twitter to predict customer engagement. Furthermore, proposing future CRM strategies by suggesting appropriate service elements for different airline companies	• This study extracted unique topics for customer needs, such as emotional experience, non-stop service, kids-related service, and service upgrade that has not been captured through the structured satisfaction survey • United Airlines, Alaska Airlines, and Delta Airlines have positive sentiments, whereas Southwest Airlines and Hawaiian Airlines have negative sentiments	680,410, airline company related tweets	Yes

ideas (Timoshenko & Hauser, 2019) to measuring readers' reactions to publicity (Park et al., 2019), many researchers have sought to apply machine learning techniques to leverage efficiency in marketing and information management research.

Prior research has highlighted the potential for social big data analytics and artificial intelligence in service management, information management, and marketing domains (see Kumar et al., 2021; Sharma et al., 2021; Verma et al., 2021). Yet most research has focused only on how to describe accurately what customers say in online environments. Critical questions remain regarding how to estimate accurately the effects of certain Twitter buzz on user engagement (See Tables 1–3).

In line with this research stream, which especially follows Kar and Dwivedi's (2020) suggestion for theory building in information management, we offer three main contributions to the literature by combining unsupervised and supervised learning processes to estimate the effects of Twitter buzz, which we defined as voluntarily posted, user-generated, word-of-mouth messages on Twitter (Clemons et al., 2006; Godes & Mayzlin, 2009; Luo & Zhang, 2013), on user engagement for different airline companies. First, by adopting the electronic word-of-mouth (eWOM) perspective as an overarching theoretical framework (Hennig-Thurau et al., 2004; King et al., 2014), we extended the concept of eWOM by empirically collecting and analyzing a large amount of tweet data (680,410 tweets collected and 4.961 tweets analyzed in the machine learning model). Specifically, previous literature on eWOM has suggested that the antecedents of a positive reaction toward eWOM are interactivity (e.g., Sheng, 2019), emotional value (Previte et al., 2019), source similarity (e.g., Hautz et al., 2014) and the valence of the eWOM message (e.g., Wakefield & Wakefield, 2018). We added sev-

Table 2
Relevant literature for social CRM.

Author (Year)	Context	Social media	Data	Relevant findings
Trainor et al. (2014)	Defining social CRM capability and investigating the relationship between customer-centric management systems, social media technology use, and social CRM that ultimately predicts customer relationship performance	NA	Survey of 308 members of top management teams	Both customer-centric management systems and social media technologies have an interactive effect on the formation of a firm-level capability that ultimately influences customer relationship performance
Tirunillai and Tellis (2014)	Customer needs extraction by utilizing online customer reviews	Customer review	350,000 consumer reviews for personal computing, mobile phones, footwear, toy, and data storage	A few dimensions, such as ease of use, secondary features, performance, visual appeal, reliability, and customer service for mobile phones, are enough to capture quality with good face validity and external validity
Misirlis and Vlachopoulou (2018)	Classifying 52 social media metrics and analytics-related articles in terms of methodology, type of analysis, field of study, marketing objectives, and social media type/ platform used	NA	NA	Natural Language Process (NNP) research consists of 15.3%, and relationship marketing (CRM and social CRM) research consists of 9.6% of 52 articles
Martí Bigorra et al. (2020)	Adopting semi-autonomous text analysis methodology for capturing customer needs for the car climate system	NA	1300 customer survey	By analyzing customer sentences, the proposed methodology helps companies to validate and update the customer needs for car climate system with an average of 90% precision and 60% recall
Harrigan et al. (2020)	Proposing social CRM framework	NA	2955 senior marketing managers in the UK	Social CRM is different from traditional CRM
Higham et al. (2021)	Identifying influencers on Twitter	Twitter	556,150 tweets	Market mavens on Twitter have more followers, post more often, have fewer readable posts, use more uppercase letters, use fewer distinct words, and use hashtags more often

Table 3
Relevant Literature on Information Management.

Author (Year)	Context	Key outcomes	Data	Sentiment Analysis	Causal Inference
Zhong (2023)	Twitter response to health news stories	The low-rated health news stories by experts are discussed more, persist longer, and produce stronger sentiments than highly-rated ones	50,000 tweets regarding health news	Yes	No
Ge et al. (2022)	Job posting analysis for Blockchain related recruitment	Blockchain development and cryptocurrency-related skill sets are the most important factors influencing salary levels	18,028 job postings from a Chinese recruitment webpage	No	Yes
Yuan et al. (2021)	Disaster resilience during Hurricane Florence focusing on demographics	The black group pays the least attention to "hurricane warn" and shows the most interest in "pray/donate," whereas the white group is most optimistic about hurricane/flood impact	2604,154 tweets	Yes	No
Pröllochs and Feuerriegel (2020)	Corporate challenge extraction through topic modeling	Based on the text data, this research quantifies the use of language into two dimensions: risk and optimism that, finally utilized to reveal the strengths and weaknesses of energy firms	29,864 8-k filing from the financial disclosures for the energy companies	No	No
Jeong et al. (2019)	Social media data analytics for product planning	Product improvement using customer comments from Reddit for Samsung Galaxy Note 5	23,614 documents	Yes	No
Son et al. (2019)	Content characteristics for effective communication strategy during the disaster	Adding more URLs to tweets increases the average retweet time more in risk-related tweets than it does in crisis-related tweets, and including key disaster-related hashtags in tweets facilitated faster retweets in crisis-related tweets.	233,928 tweets during the 2013 Colorado flood.	No	No
Palese and Usai (2018)	Service quality importance	The distribution of topics in online opinions differs between positive and negative reviews, and concerns about merchant responsiveness are the most important dimensions for negative reviews	74,775 online reviews from a leading online price comparison company in Italy	No	Yes
This Study	Identifying factors affecting customer engagement in Twitter for airline companies and estimating the effects of different topics discussed on Twitter to predict customer engagement. Furthermore, proposing future CRM strategies by suggesting appropriate service elements for different airline companies	• This study extracted unique topics for customer needs, such as emotional experience, non-stop service, kids-related service, and service upgrade that has not been captured through the structured satisfaction survey • United Airlines, Alaska Airlines, and Delta Airlines have positive sentiments, whereas Southwest Airlines and Hawaiian Airlines have negative sentiments.	680,410, airline company related tweets	Yes, utilizing machine learning	Yes, utilizing Poisson Regression

eral dimensions of linguistic features such as the length of the word, the sentiment of a tweet, the gap between the date the tweet was generated and the retweeted date, and—most importantly—the different themes of the tweets for each airline company. Second, we conducted a more rigorous unsupervised machine procedure called latent Dirichlet allocation to generate the latent topics for the user-generated content (UGC) related to the airline company in the Twitter environment (LDA; Blei et al., 2003). Third, most interestingly, we found the associations between generated features (i.e., independent variables of prominent topics) to consumer engagement (i.e., the number of retweets and number of likes) by adopting a rigorous estimation technique such as Poisson regression. Kar and Dwivedi (2020), in fact, highlighted the limited research in this area and the opportunities for future research on how user-generated content could be mined to explain user behavior. Inspired by the potential of Twitter buzz for capturing customers' needs and generating customer engagement, we considered the following three research questions.

- **RQ1**: What is the *prominent topic* of Twitter buzz for various airlines, and how do the topics *differ* across the airline companies? That is, what are the top five topics extracted from the Twitter buzz for United Airlines, Delta Airlines, Southwest Airlines, Alaska Airlines, and Hawaiian Airlines?
- **RQ2**: What is the *valence* of the Twitter buzz for various airlines, and how does the sentiment *differ* across the airline companies? That is, to what extent is Twitter buzz positive or negative for United Airlines, Delta Airlines, Southwest Airlines, Alaska Airlines, and Hawaiian Airlines?
- **RQ3**: What is the effect of Twitter buzz on consumer engagement for various airlines, and how do the effects of Twitter buzz *differ* across the airline companies? That is, how does the gap between the tweet generated date and the retweeted date, buzz sentiment, the number of hashtags, the number of words used in the tweet, and the effects of each topic influence users' engagement on Twitter (i.e., the number of retweets and number of likes)?

By answering these research questions, this study ultimately explained why customers reacted differently based on the textual features and demonstrated how to accurately extract customer needs through the Twitter buzz that ultimately influences user engagement.

The following sections discuss this study's conceptual background by integrating the literature streams of the effects of UGC and artificial intelligence (AI) using the eWOM framework. The feature-generating methodology, estimation procedure, and results of statistical modeling will then be presented. Finally, the limitations and strategic recommendations for developing theory and improving information management practice will be discussed.

2. Related works

2.1. Airline industry management and social customer relationship management (Social CRM)

Our study built on airline industry management and social CRM literature as summarized in Tables 1 and 2. Pioneering research in this area demonstrated the importance of listening to customers' voices to manage the customer relationship (Kumar et al., 2016; Malthouse et al., 2013). Social CRM refers to the adaptation of social media to allow the creation and exchange of user-generated content for relationship initiation (acquisition), maintenance (retention), and termination (Kaplan & Haenlein, 2010; Malthouse et al., 2013). Trainor et al. (2014) defined social CRM as "the integration of traditional customer-facing activities, including processes, systems, and technologies, with emergent social media applications to engage customers in collaborative conversations and enhance customer relationships" (Trainor et al., 2014, p. 1202). However, a majority of previous research in airline industry management utilized traditional approaches such as surveys (e.g., Chen et al., 2019; Ringle et al., 2011; Zhang et al., 2019) and case studies (e.g., Palma-Mendoza et al., 2014). For example, Ringle et al. (2011) found that perceived safety is one of the most important predictors of airline customer satisfaction level, and Zhang et al. (2019) further extended the airline industry CRM into specific airline related products such as aviation voluntary carbon offsetting (VCO) products. Chen et al. (2019) revealed the factors affecting mobile app adoption for airline customers.

Notably, most studies to date have focused on CRM strategy for the airline industry either utilizing self-reported survey methodology or qualitative research methods such as interviews or case studies. Although this traditional methodology revealed the underlying dynamics of airline customers' satisfaction levels with the service, customers' opinions from the structured survey might differ from their opinions on unstructured data such as online reviews (Tirunillai & Tellis, 2014). For example, Tirunillai and Tellis (2014) analyzed customer reviews related to mobile phones and found that messaging, exhaustibility (battery), and layout discomfort are important dimensions for consumers to consider mobile phone purchases, which were not captured through structured surveys (i.e., consumer reports). Therefore, we analyzed buzz from Twitter to capture the unique CRM dimensions to satisfy customers' needs that were expressed in social media (Harrigan et al., 2020; Wang & Kim, 2017).

2.2. Information management for big data analysis

As Sharma et al. (2021) illustrated, data mining, especially big data analysis, is one of the most important research areas in the information management field. Several studies have investigated the potential of social big data to extract insights in a variety of contexts, including service quality importance extraction (Palese & Usai, 2018), disaster communication strategy building (Ahn et al., 2021; Son et al., 2019; Yuan et al., 2021), and product planning (Jeong et al., 2019). For example, Son et al. (2019) found that adding more URLs to tweets increased the retweet time during the 2013 Colorado flood. Ahn et al. (2021) observed that tweets from non-media outlets generated more engagement during the 2019 Ridgecrest earthquake in Southern California. More recently, Zhong (2023) found that low-rated health news stories by experts are circulated more, persist longer, and produce stronger sentiments in the Twitter environment in terms of health news stories. Ge et al. (2022) extracted the job postings regarding Blockchain in China and found that Blockchain development and cryptocurrency-related skill sets are the most important factors in determining salary level. As the list in Table 3 demonstrated, however, the majority of information management literature has relied on supervised learning process such as regression (e.g., Palese & Usai, 2018), sentiment analysis (e.g., Yuan et al., 2021; Zhong, 2023), or the unsupervised learning process of topic modeling (e.g., Ge et al., 2022; Jeong et al., 2019; Pröllochs & Feuerriegel, 2020; Son et al., 2019).

In light of utilizing text mining in the tourism research domain, it is noteworthy that several information systems research guide the current study (e.g., Agrawal et al., 2022; Altuntas et al., 2022; Grover & Kar, 2017; Kar et al., 2021; Kushwaha et al., 2021; Mishra et al., 2023; Obembe et al., 2021). For example, in the context of service experience that might be applied to tourism satisfaction, Kar et al. (2021) found that accessibility, accommodation, assurance, cultural attraction, Jugaadu service flexibility, cleanliness, hospitality, price, restaurant, and security were the predictor of service experience using topic modeling. Particularly focusing on the tourism research context, Agrawal et al. (2022) proposed the potential utilization of big data analytics in sustainable tourism research, and Altuntas et al. (2022) analyzed the quarantine decision during the COVID-19 pandemic that influenced tourism in Türkiye. Similarly, Obembe et al. (2021) analyzed the impact of social media during the COVID-19 crisis on the tourism industry. Mishra et al. (2023) utilized clustering and topic modeling to identify the topic from reviews to predict customers' reactions toward tourism promotions ultimately.

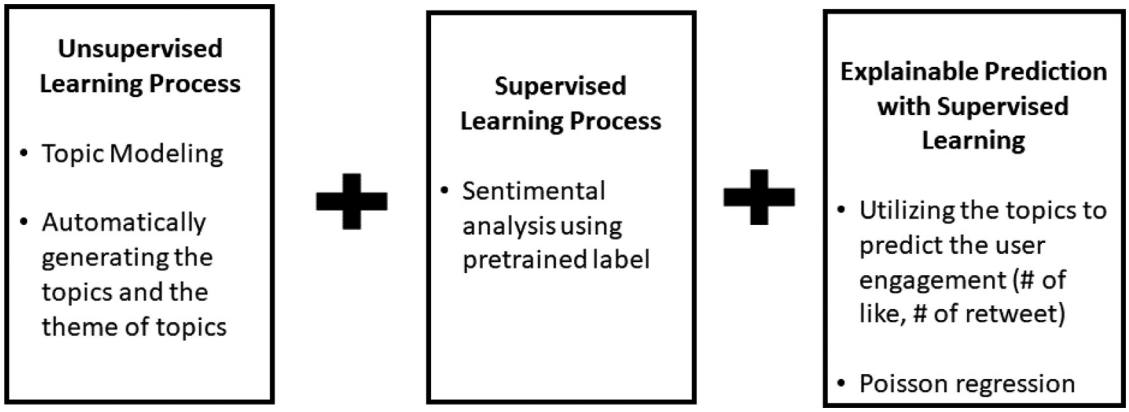

Fig. 1. Study overview.

We drew on these foundations to develop a novel approach to capture the needs of customers of the different airline companies by integrating supervised and unsupervised learning methods simultaneously.

2.3. eWOM and UGC

As an overarching framework, we adopted the eWOM perspective, which is defined as "any positive or negative statement made by potential, actual, or former customers about a product or company, which is made available to a multitude of people and institutions via the Internet" (Hennig-Thurau et al., 2004, p. 39). Researchers have expanded the concept of eWOM based on the original definition of word-of-mouth as an informal, non-commercial flow of information primarily regarding post-purchase behavior (Herr et al., 1991). In this study, we considered a variety of types of UGCs including tweets (Aleti et al., 2019) and customer reviews (Chevalier & Mayzlin, 2006; Tirunillai & Tellis, 2014). We argued that messages in the social media environment, such as tweets on Twitter, can also be considered eWOM messages since most tweets represent informal processes among consumers.

By adopting this basic definition, previous literature has investigated the antecedents of eWOM (Gottschalk & Mafael, 2017) and their effects on consumers' responses including attitude toward the brand, purchase intention, and sales (Chevalier & Mayzlin, 2006; Gopinath et al., 2013, 2014). For example, Gottschalk and Mafael (2017) evaluated the eWOM processing types to classify the antecedents of eWOM behavior.

Previte et al. (2019) identified that emotional value is an important antecedent of users' engagement in eWOM behavior. Wakefield and Wakefield (2018) investigated the relationship between anxiety, message availability, and negative eWOM creation. Notably, Aleit et al. (2019) evaluated celebrity social media communication in Twitter to analyze consumers' eWOM behavior. They found that narrative styles were positively related to the users' eWOM behavior, whereas emotional styles were not.

Considering an individual consumer's tweet to be a form of eWOM, we proposed and conceptualized novel antecedents of eWOM effectiveness: timing, word count, the number of hashtags, the sentiment of the tweet, and the prominent topics of Twitter buzz. Therefore, we aimed to theoretically conceptualize and empirically demonstrate the underlying eWOM mechanism.

3. Methodology

3.1. Using artificial intelligence (AI) to analyze unstructured data

A variety of methods have been used to extract and analyze consumer opinions (e.g., Babbar & Koufteros, 2008; Ringle et al., 2011). For example, Babbar and Koufteros (2008) analyzed 437 questionnaires and found that individual attention, helpfulness, courtesy, and promptness collectively were the positive antecedents of individual passengers' satisfaction level. Ringle et al. (2011) pointed out that perceived safety is an important construct for customers' level of satisfaction with the aircraft experience during their travels. Experiments have also been used to test the effects of different antecedents on airline service, and Hazée et al. (2017) found that a co-created suggestion is effective when passengers have experienced service failure.

In this study, we adopted cutting-edge methodology to analyze a large amount of text data by using the unsupervised learning process of LDA from the artificial intelligence (AI) field. Historically, we can position the AI definition into four categories based on the "rationality vs. humanity" dimension on the x-axis and the "thinking vs. acting" dimension on the y-axis (Russell & Norvig, 2010). For example, Haugeland (1985) defined AI as machines with minds by following the "thinking humanly" dimension. Rich and Knight (1991) referred to AI as the study of how to make computers do things that, at the moment, people do better by focusing on the "acting humanly" dimension. Based on the "thinking rationally" aspect, Winston (1992) defined AI as the study of the computations to make this computation possible to perceive, reason, and act. Finally, Poole et al. (1998) defined AI as a computational intelligence, or the designing of intelligent agents rooted in the "acting rationally" dimension. Based on these basic definitions of AI, today's researchers refer to AI as a broad field of science that utilizes data sets by continuously changing optimized outcomes from the data to figure out the most appropriate solution (Davenport et al., 2017). Thus, a modern definition of AI contains the possible utilization of big data by focusing not only on the computer science field, but also psychology, linguistics, philosophy, and other areas (van Duin & Bakhshi, 2018).

Therefore, we focused especially on the modern definition of AI's "learning" process that continuously changes and adopts the optimized outcome from the data to model the optimal solution, which can be classified into a machine learning process for AI (see Fig. 1). Indeed, this learning process is a key function of AI and, through the learning process, AI can learn how to think, learn humanly, and reflect the rationality of human beings. Indeed, several researchers have investigated using AI in different settings such as e-learning (Almaiah et al., 2022a, 2022b) and marketing (Verma et al., 2021).

Researchers from the engineering and statistics domain classified the learning process into unsupervised and supervised learning processes based on the feedback. Unsupervised learning is a type of machine learning procedure that utilizes a dataset without prelabels to recognize the patterns in the dataset (Hinton et al., 1999), whereas supervised learning utilizes prelabeled input–output pairs (Russell & Norvig, 2010). Statistically, supervised learning employs training data comprising examples of the input vectors corresponding to the target vectors (Bishop, 2006). For example, the classical digit recognition example that aims to assign each input vector to one of the finite number of categories (e.g., assign image data to category 1 or 2) is a supervised learning process. In contrast,

unsupervised learning employs input vectors without any corresponding target values (Bishop, 2006). In this learning process, we extracted similar patterns from within the data. For example, we figured out the similarities among respondents' characteristics and classified groups of similar examples into the clustering analysis.

By adopting the definition of AI and different types of learning in this study, we employed both the unsupervised learning process for generating and the supervised learning process for estimating the effects of features on dependent variables. In the following section, we illustrate the unsupervised and supervised learning processes used in this study in detail.

3.2. Data collection method

3.2.1. Data source: Twitter

Our primary goal was to use UGC to predict users' engagement levels (i.e., the number of retweets and the number of likes for airline companies). We particularly selected Twitter as the context for this study context for mainly two reasons: First, a recent meta-analysis indicated Twitter is more effective for generating stronger positive satisfaction and positive emotion than Facebook or blogs in terms of customer engagement (de Oliveira Santini et al., 2020). Second, unlike other social media platforms such as Facebook or blogs, Twitter is a particularly useful tool for social CRM (Harrigan et al., 2021, 2020; Pournarakis et al., 2017; Risius & Beck, 2015). Specifically, Twitter is the second most popular social media platform for social CRM, followed by LinkedIn Harrigan et al. (2020), and the hashtag function of Twitter enables researchers and marketers to search, download, and analyze public opinion utilizing "big data" conveniently (Harrigan et al., 2021; Kumar et al., 2021; Risius & Beck, 2015). In addition, Twitter's Application Program Interface (API) enables researchers to collect a targeted large amount of publicly available tweets that are mostly unbiased (Harrigan et al., 2021; Risius & Beck, 2015). In the following sections, we explain how we collected the data, cleaned the data, and defined the descriptive statistics for our dataset.

3.2.2. Data collection procedure

The data in this study were collected from Twitter (www.twitter.com) using Twitter's streaming APIs and Python's Tweepy library. Streaming APIs push the tweet or retweet to the end point if it includes a specific keyword. We collected tweets containing any of the following keywords: "United Airlines," "American Airlines," "Delta Airlines," "Virgin Atlantic," "Alaska Airlines," "Hawaiian Airlines," "Spirit Airlines," or "Southwest Airlines," which were representative airlines in the United States from May 1 to August 4, 2016 (95 days). During this period, a total of 680,410 tweets and retweets that included the keywords were further filtered using regular expression and Python's Pandas library functions.

3.2.3. Data cleaning and transformation

We first removed any duplicate tweets and retweets from the 680,410 tweets. When Twitter streaming APIs push the tweet and retweet to the end point, a tweet retweeted by several users is also pushed. Thus, having duplicates in the model may introduce noise into the model, and they should be removed. The authors kept only the last retweeted message in the collected tweets and retweets in order to clean the data. For example, "Muslim family asked to step off United Airlines plane for 'flight safety' issue. #discrimination" was the most retweeted tweet (1376 times) during the data collection period with the keyword "United Airlines." In this case, we recorded 1376 retweets for this individual tweet to utilize as a dependent variable but eliminated the other 1376 tweets for the topic modeling and other analysis procedures. In addition, to model accurately the effects of independent variables on user engagement (e.g., retweets and likes), we excluded the tweets that were never forwarded to other users (i.e., the number of retweets is 0) and left the retweeted tweets.

Using the remaining retweeted texts, we cleaned the text data using the following steps:

1. Removed the http, https, and URLs representing outside source links from the tweet text
2. Removed hashtags used for the keyword search in Twitter
3. Removed user name(s) mentioned in the text, retweet symbols (RT @), and retweeted user name
4. Removed non-alphabetic words
5. Replaced capital letters with lower-case letters and removed stop words such as "a" and "the" using the standard vocabulary of stop words in the English language

Using this procedure, we found that tweets containing "American Airlines," "Virgin Atlantic," and "Spirit Airlines" contained a lot of noisy words, so we excluded these tweets from our analysis. Thus, our final dataset contained a total of 4961 tweets (United Airlines: 1591, Delta Airlines: 843, Southwest Airlines: 1771, Alaska Airlines: 489, and Hawaiian Airlines: 267).

3.3. Feature generation process

3.3.1. Unsupervised learning: LDA

Among various text data analysis methods (Kumar et al., 2021), we particularly employed Latent Dirichlet Allocation (LDA, Blei et al., 2003) since this is one of the most useful techniques to capture customers' needs based on the social CRM framework (Ahn et al., 2021; Kumar et al., 2021). Our data analysis included two approaches: topic modeling for generating features (i.e., independent variables) and Poisson regression to estimate the effects of independent variables on user engagement level in Twitter. Using Python's scikit-learn package, LDA (Blei et al., 2003) was implemented as a method of topic modeling. The LDA assumes the existence of a fixed number of latent topics among the multiple documents (e.g., tweets) and analyzes and inferences the latent topic based on the co-occurring probability.

Following Blei et al.' (2003) basic notation, we assumed the generative process for each document **w** in a corpus D:

1. Choose N$\sim$ Poisson (ξ).
2. Choose $\theta \sim$ Dirichlet (α).
3. For each of the N words w_n:

   (a) Choose a topic $z_n \sim$ Multinomial (θ).
   (b) Choose a word w_n from $p(w_n|z_n,\ \beta)$, a multinomial probability conditioned on the topic z_n.

In this study context, each tweet is described by its own mixture of topics. Thus, a *word* is the discrete words in each tweet, defined as an item from the vocabulary indexed by $\{1, \ldots, V\}$. A *document* is a sequence of N words, which is a single tweet. This document is denoted by $\boldsymbol{w} = (w_1, w_2, \ldots, w_N)$, where w_n is the nth word in the sequence. A corpus is a collection of M documents denoted by $D = \{\boldsymbol{w}_1, \boldsymbol{w}_2, \ldots, \boldsymbol{w}_M\}$. Therefore, we can denote the document-specific probability vector associated with the topics z_{dn}as θ_d. Also, β is a matrix of word–topic probabilities $\{\beta_{m,t}\}$ for word m and topic t, with $p(w_{dn} = m|z_{dn} = t,\ \beta) = p(w_{dn} = m|\ \beta_t)$. In this unsupervised learning process, the ultimate goal of the machine learning process is estimating, β_t, a vector of word probabilities for topic t. In particular, we selected LDA for the unsupervised learning procedure as the specification of the LDA model assumes only a homogenous Dirichlet prior for the θ_d and β_t and does not impose a "bag-of-words" assumption in the text analysis literature. Therefore, we can expand our estimation without the assumption that the latent topic z_{dn}can vary from word to word (Blei et al., 2003; Büschken & Allenby, 2016).

Based on the assumptions of the basic model, we inferred the hidden variables by computing their distribution conditioned on the documents: p(topics, proportions, assignments |documents). Therefore, we computed

the distribution as follows:

$$p(\beta,\ \theta, \boldsymbol{z},\ \boldsymbol{w}) = \left(\prod_{i=1}^{k} p(\beta_k|\eta)\right)\left(\prod_{d=1}^{D} p(\theta_d|\alpha)\prod_{n=1}^{N} p(z_{dn}|\theta_d)p(w_{dn}|\beta_{1k}, z_{dn})\right)$$

This joint defines a posterior as $p(\theta,\ z, \beta \mid w)$, where θ is a distribution of topics, z is a set of N topics, β is a distribution of words (one for each topic), and w is all data (i.e., all the tweets in the model). Thus, z_{dn} is a pre-word topic assignment, θ_d is a pre-document topic proportion, and $\beta_k$is a pre-corpus topic distribution while α is a parameter vector and η is a parameter vector for each topic.

To infer an approximate posterior, previous literature has suggested a variety of inference algorithms including mean field variational methods (Blei et al., 2003), expectation propagation (Minka & Lafferty, 2002), collapsed Gibbs sampling (Griffiths & Steyvers, 2004), distributed sampling (Newman et al., 2008), collapsed variational inference (Teh et al., 2008), factorization-based inference (Anandkumar et al., 2012; Arora et al., 2012), and online variational inference (Hoffman et al., 2010). Among these suggested algorithms, we specifically adopted the online variational Bayes (VB) algorithms (Hoffman et al., 2010) as this algorithm significantly reduces the memory requirement for computation and converges faster than other algorithms (Hoffman et al., 2010). Based on the Bayesian paradigm, VB inference approximates the true posterior by indexing a set of free parameters of $q(z,\ \theta, \beta)$ to optimize by maximizing the Evidence Lower Bound (ELBO).

After conducting an initial LDA analysis, we estimated the optimal number of topics by checking both the perplexity score and coherence score of LDA models. Previous literature indicated that we should consider several key questions for determining the optimal number of topics (Stevens et al., 2012). 1) How many topics should be learned? 2) How many learned topics are useful? 3) How do these topics relate to often used semantic tests? 4) How well do these topics identify similar documents?

First, we calculated the coherence score specifically utilizing the UMass metric. Among several coherence score metrics, such as the UCI metric and C_V metric, the UMass metric is generally considered the most appropriate methodology for calculating the LDA model's coherence score since this metric is most intrinsic in nature to learn about data in the collected corpus (Stevens et al., 2012). The UMass metric generates the score based on the document co-occurrence, where $D(x,\ y)$ is the frequency of the number of documents containing words $x,\ y\ and\ D(x,\ y)$ represents the number of documents containing x. Therefore, the UMass score is defined as follows (Stevens et al., 2012):

$$Coherence\ score\ \left(v_i, v_j, \in\right) = \log \frac{D\left(v_i,\ v_j\right) + \in}{D\left(v_j\right)}$$

The statistical criterion for selecting an optimal number of topics is selecting the lowest score in coherence score. We varied the number of topics starting from 1 to 30 and calculated the coherence score and compared each score (Fig. 2). As Fig. 2 indicates, as we increased the number of topics, the coherence score dropped. However, it is unrealistic to employ a large number of topics for building CRM strategies and future estimation processes.

Similarly, we also calculated the document perplexity, which focuses on the log-likelihood of a hold-out test set as flows (Wallach et al., 2009):

$$L(\omega) = \log p(\omega|\Phi, \alpha) = \sum_{d} \log p(\omega_d|\Phi, \alpha)$$

where the collection of unseen documents is ω_d, the topic matrix is Φ, and the hyper parameter for the topic distribution of a document is α. Therefore, the measure of perplexity is defined as:

$$Perplexcity(\omega) = exp\left\{-\frac{L(\omega)}{count\ of\ tokens}\right\}.$$

Therefore, we decided on five topics by analyzing coherence and perplexity scores (perplexity score: −8.8841, coherence score: −10.1100) since when we set the number of topics as five, it showed a sharp decrease in both perplexity and coherence scores. Through these processes, we could confirm the validity and reliability of topic modeling (Table 4).

Table 4
The number of topics, perplexity score and coherence score.

# of Topics	Perplexity	Change of Perplexity Score	Coherence	Change of Coherence Score
1	−8.4081		−7.7461	
2	−8.5159	.1078	−9.9330	2.187
3	−8.6361	.1202	−8.9833	.9497
4	−8.7529	.1168	−9.0253	.0420
5	**−8.8841**	**.1312**	**−10.1100**	**1.0847**
6	−9.0220	.1379	−10.1216	.0116
7	−9.1540	.1320	−11.1237	1.0021
8	−9.2896	.1356	−11.9124	.07887

3.3.2. *Supervised learning: sentiment analysis*

After generating topics for the tweets, we analyzed the sentiment of each tweet to define one of our main independent variables. We adopted the Valence Aware Dictionary for sEntiment Reasoning (VADER; Hutto & Gilbert, 2014) in order to quantify accurately each tweet's sentiment. Although previous dictionary-based sentiment analysis methods, such as Linguistic Inquiry and Word Count (LIWC; Pennebaker et al., 2007), have provided insights for the automatic content analysis, VADER provides several advantages over other dictionary-based sentimental analysis techniques based on the word frequency approach for mainly four reasons. First, VADER's lexicon was developed using a combination of qualitative and quantitative methods. Second, previous sentiment analysis tools such as LIWC were developed to analyze longer sentences, paragraphs, or even books, and thus, they have limited performance for analyzing short sentences such as tweets (maximum 140 words). Third, most previous sentimental analyses are unable to differentiate the intensity of the sentiment. For example, "The food here is *excellent*" expresses more positive intensity than "The food here is *okay*," but LIWC would give the same positive score of "1″ since they both contain only one positive word (exceptional and okay). By developing pretrained label data with Amazon MTurk's rating of more than 9000 lexical feature candidates, VADER differentiates the intensity of positive emotion by giving 2.7 points for *excellent* and 0.9 points for *okay*. Finally, the accuracy for predicting sentiment when using VADER (F1 score for the accuracy of classification = 0.96) is higher than that of human coders (F1 = 0.84).

Therefore, we used the VADER algorithm as a supervised learning process to evaluate each tweet's sentiment and numerically quantify the sentiment score based on three categories: positive versus negative versus neutral, not only considering the binary polarity but also calculating the intensity (i.e., strength) of the sentiment expressed in the tweets. Specifically, the VADER algorithm calculates the valence scores of each word in the text data based on the pre-labeled lexicon and displays the probability of whether each tweet can be classified as a positive versus negative versus neutral sentiment (Hutto & Gilbert, 2014). In addition, the VADER algorithm allows for calculating the "compound score" as follows: $Compound = \frac{\sum i\ Sentiment\ (w_i)}{\sqrt{\sum iSentiment(w_i)^2 + \alpha}}$, where *sentiment(w)* is the score valance of the word w and α is a normalization coefficient maximizing the expected value (Bonaccorso, 2018). By calculating this compound score, we can accurately estimate the sentiment for each tweet. Table 5 shows the variables' conceptual definition and the generating method in our analysis.

3.4. *Data analysis method*

3.4.1. *Estimating the effects of features on the dependent variables*

We investigated the use of latent topics and other features to predict and explain users' engagement level on Twitter. The purpose of our

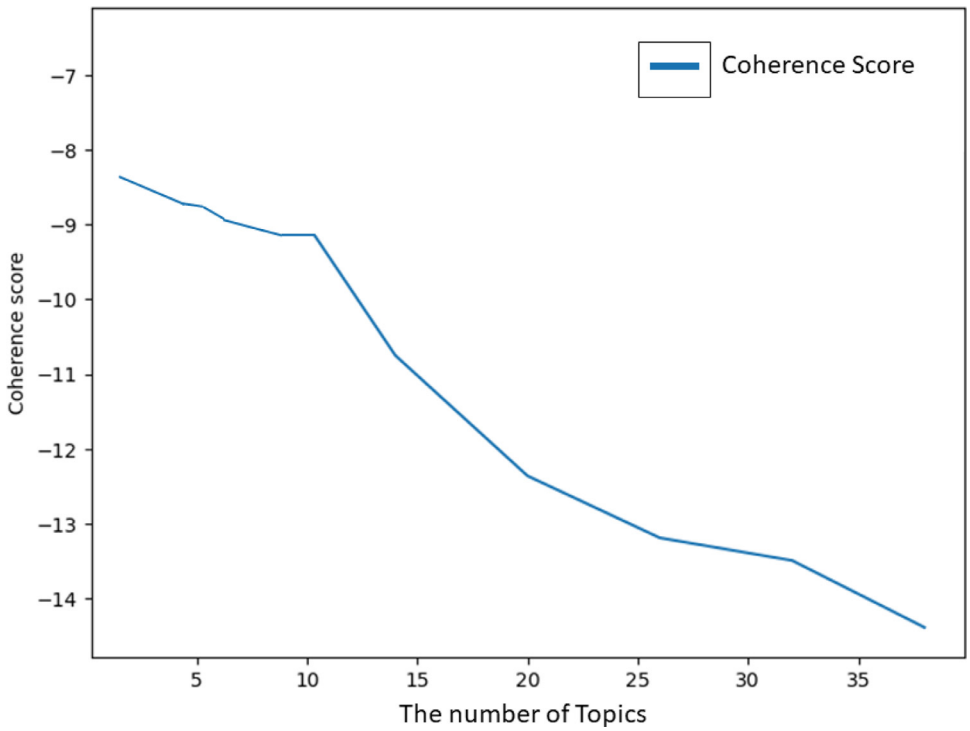

Fig. 2. Coherence score.

Table 5
Variables in the Poisson regression with description and generating method.

Variable	Description	Operationalization and Generating Method
Tweet_RT gap	Distance between the original tweet generated day and the retweeted day (day). Since we included only retweeted tweets in our analysis, the minimum value is 0 (no time gaps between the original tweet generated day and the retweeted day)	Subtracting the time when the original message was published from the time when the last retweet was made from
Compound	The sentiment level of each tweet. −1 represents extremely negative and +1 represents extremely positive (−1 < compound < +1).	Supervised machine learning: VADER
HashtagCount	The number of hashtags in each tweet	Count
WordCount	The number of words in each tweet	Count
Topic factor	Among five topics, assigned topic per each tweet based on the word probability (dummy coded)	Unsupervised machine learning: LDA
Number of retweets	The primary measure of engagement level. The number of retweets each tweet received.	Count
Number of likes	The primary measure of engagement level. The number likes each tweet received.	Count

analysis was to utilize extracted themes and other features, such as word count, sentiment of tweets, number of hashtags, and tweet–retweet gap for predicting users' number of likes and retweets.

3.4.2. Supervised learning: predicting user engagement using Poisson regression

As a supervised learning process for estimating the effects of features on the dependent variables, we selected a Poisson regression (i.e., a generalized logistic regression employed to analyze count Faraway, 2016; Long & Freese, 2006) since our main dependent variables were count data (i.e., the number of retweets and the number of likes). In addition, dependent variables in our data contained many numerical values of zeros and ones ($N_{retweet}$ = 2557, 52.34%; N_{like} = 2463, 54.33%) and were highly skewed (Retweet skewness: 26.25, kurtosis: 947.45; Like skewness: 14.648, kurtosis: 274.34) with non-normal distribution. The Poisson regression can incorporate values of zero in the dependent variables by maintaining a log-linear form (Fisher et al., 2019), which allows for accurately estimating the effects of the independent variables on the number of retweets and the number of likes that include zero values.

Therefore, Poisson regression provided a much more valid approach to model our data than the ordinary least squares (OLS) regression. Indeed, previous studies that analyzed count data, such as the number of helpful votes as a dependent variable, also employed the Poisson regression for their estimations (Kupor & Tormala, 2018). From the information management literature perspective, we observed the growing number of research studies that have utilized Poisson distribution to estimate the effects of independent variables on dependent variables (e.g., Barati & Yankson, 2022; Kutela et al., 2022; Lukinskiy et al., 2023; Zhang et al., 2022). For example, Barati and Yankson (2022) used the Poisson distribution to predict data breach occurrence, and Kutela et al. (2022) analyzed social media followers' engagement using negative binomial regression, which is a similar approach. Zhang et al. (2022) also estimated

Table 6
Summary statistics.

	United Airlines	Delta Air Lines	Southwest Airlines	Alaska Airlines	Hawaiian Airlines
Number of Tweets	1591	843	1771	489	267
Word Count					
Mean (SD)	15.37 (4.48)	15.48(5.03)	13.69(4.27)	14.35(3.93)	13.10(4.01)
Median	15	16	13	14	13
Range	3 – 28	3 – 30	2 – 28	3– 27	3– 27
Number of Like					
Mean (SD)	6.34(40.49)	6.38(28.47)	5.94(32.06)	3.77(12.38)	4.33(16.58)
Median	1	0	0	1	1
Range	0 – 723	0 – 377	0 – 782	0 – 163	0 – 221
Number of Retweets					
Mean (SD)	6.13(39.12)	5.68(19.28)	5.68(24.77)	5.95(48.12)	4.26(6.89)
Median	1	1	1	1	2
Range	1 – 1376	1 – 221	1 – 344	1 – 1043	1 – 52
Number of Mentions					
Mean (SD)	.35(0.65)	.26(0.65)	.22(0.50)	.27(0.60)	.19(0.55)
Median	0	0	0	0	0
Range	0 – 5	0 – 7	0 – 4	0 – 4	0 – 3
Number of Hashtags (SD)					
Mean (SD)	4.14(8.45)	3.50(7.70)	3.92(8.26)	4.49(8.70)	5.26(8.53)
Median	0	0	0	0	0
Range	0 – 65	0 – 56	0 – 73	0 – 60	0 – 46
Tweet-retweet gap (day)					
Mean (SD)	2.96(17.31)	4.06(23.13)	2.06(10.18)	2.59(14.29)	2.75(16.99)
Median	0	0	0	0	0
Range	0 – 250	0 – 269	0 – 149	0 – 186	0 – 204
Mean Sentiment Score of Tweets (SD): (−1 < sentiment < 1)					
Positive	.34(0.15)	.12(0.18)	.07(0.14)	.11(0.16)	.06(0.12)
Negative	−.08(0.14)	−.09(0.15)	−.09(0.19)	−.03(0.09)	.07(0.13)
Neutral	.57(0.16)	.80(0.02)	.82(0.19)	.86(0.17)	.87(0.17)
Compound	.37(0.37)	.06(0.41)	−.03(0.35)	.13(0.30)	−.03(0.29)

the effects of different topics in social media to predict customers' engagement levels using Poisson regression.

Specifically, our Poisson regression model assumed that the outcome variable of the number of retweets and the number of likes followed a Poisson distribution. If y had a Poisson distribution, we modeled the dependent variable as a probabilistic model: $\Pr(y \mid \mu) = \frac{e^{-\mu}\mu^{y}}{y!}$ for $y = 0$, 1, 2, …, and we followed the basic assumption of Poisson regression with the mean of μ_i, where each observation of y had a different value of μ. Thus, we estimated the effects of the independent variables on the dependent variables as follows:

$$\mu_i = (\beta_0 + \beta_1 Twet_RT_Gap + \beta_2 Compound + \beta_2 Hashtag\ Count + \beta_3 Word\ Count + \beta_4 Topic\ 1 \quad \beta_5 Topic\ 2 + \beta_6 Topic\ 3 + \beta_7 Topic\ 4 + \beta_8 Topic\ 5)$$

4. Results

4.1. Descriptive statistics

Before presenting the results of the statistical analysis, we report the descriptive statistics for our variables in Table 6. On average, each tweet in our dataset contained 14.40 words (*SD*: 1.04) and 4.26 hashtags (*SD*: 0.66), generating 5.54 retweets (*SD:* 0.74) and 5.35 likes (*SD*: 1.22). The average gaps between the tweet generated date and the retweeted date was 2.88 days (*SD* = 0.74). The average sentiment of tweets was 0.1 (*SD* = 0.17).

4.2. What was the prominent topic of the Twitter buzz, and how did the topics differ across the airline companies? (RQ1)

To answer research question 1, we conducted LDA for the tweets with online VB algorithms for the inference. We first extracted the top five topics for each airline company and sorted the word probability to label each topic. United Airlines' prominent topics were business-related issue, special service, airport service, delayed complaint, and flight attendant. Users mentioning Delta Airlines frequently discussed themes related to service, delayed complaint, bad flight experience, ticketing experience, and passenger experience. Southwest Airlines was associated with canceled flight, travel plan, flight delay experience, customer service, and emotional experience. Alaska Airlines was frequently mentioned within topics such as Alaska Airlines brand, flight news, offer and deal, flight experience, and non-stop service. Hawaiian Airlines was associated with the Hawaiian Airlines brand, seat and class, kids-related service, delayed complaint, flight experience, and service upgrade (Table 7).

4.3. What was the valence of the Twitter buzz and how did the sentiment differ across the airline companies? (RQ2)

To measure the valence of tweets of the Twitter buzz, we conducted a sentiment analysis employing the VADER algorithm and generated a compound score for each tweet (Hutto & Gilbert, 2014). The compound scores ranged from −1 to +1, where −1 represented an extremely negative sentiment and +1 showed an extremely positive sentiment, by considering both the sentiment of the words and the intensity of the sentiment. Among the five airline companies, tweets related to the United Airlines exhibited the highest level of positive sentiment ($M_{\text{United Airlines}} = 0.37$, $SD = 0.37$), followed by the Alaska Airlines ($M_{\text{Alaska Airlines}} = 0.13$, $SD = 0.30$) and Delta Airlines ($M_{\text{Delta Airlines}} = 0.06$, $SD = 0.41$). However, Twitter buzz related to Southwest Airlines ($M_{\text{Southwest Airlines}} = -.03$, $SD = 0.35$) and Alaska Airlines ($M_{\text{Southwest Airlines}} = -.03$, $SD = 0.29$) was negative (Table 6).

4.4. What was the effect of Twitter buzz on consumer engagement, and how did the effects of Twitter buzz differ across the airline companies? (RQ3)

Given the rapid usage of Twitter buzz to capture the unfiltered consumer response to the brand and company, previous literature has con-

Table 7
(1) Top words from the LDA model based on the probability of word occurrence ($T = 5$): United Airlines.

Topic 1: "Business Related Issue" $N = 748$ (47.01%)	Topic 2: "Special Service" $N = 165$ (10.37%)	Topic 3 "Airport Service" $N = 175$ (10.10%)	Topic 4: "Delay Complain" $N = 290$ (18.23%)	Topic5 "Flight Attendant" $N = 213$ (13.39%)
fly	travel	service	flight	make
Suspend	air	return	delay	emergency
Guilty	dog	new	land	attendant
Time	rescue	turn	hour	landing
Plead	meat	airport	cancel	class
Say	request	day	divert	business
lawmaker	trade	free	help	unite
Hang	contract	stop	issue	thank
Use	agree	announce	live	deal
Receive	trade	work	hope	ticket
Want	charge	mile	refund	leave
Update		food	team	truly
Start		strike	customer	hit
Declare		good	short	plan
Bad		trip	add	depart
response		sign	company	bind
Suck		force	big	crash
Case		amp	approve	reach
Scandal		pilot	man	
Video		let	lounge	
Swanky		bird	high	
Problem		petition	illy	
Execute		people	fare	
Public		passenger	threaten	
Fall			exclusive	
Reality			need	
Long			safely	
			happen	
			home	
			coffee	
			approval	

(2). Top words from the LDA model based on the probability of word occurrence ($T = 5$): Delta Airlines

Topic 1: "Service" $N = 357$ (60.71%)	Topic 2: "Delay Complain" $N = 54$ (9.18%)	Topic 3: "Bad Flight Experience" $N = 76$ (12.93%)	Topic 4: "Ticketing Experience" $N = 53$ (9.01%)	Topic5 "Passenger Experience" $N = 48$ (8.16%)
airline	fly	flight	ticket	divert
remove	plane	land	trip	passenger
vomitte	time	mistake	win	today
good	thank	travel	enter	pray
	make		round	
	delay			
	team			

(3). Top words from the LDA model based on the probability of word occurrence ($T = 5$): Southwest Airlines

Topic 1: "Flight Cancel" $N = 538$ (30.38%)	Topic 2: "Travel Plan" $N = 257$ (14.51%)	Topic 3: "Flight Delay Experience" $N = 350$ (19.76%)	Topic 4: "Customer Service" $N = 242$ (13.66%)	Topic5 "Emotional Experience" $N = 384$ (21.68%)
flight	fly	delay	say	airline
cancel	plane	outage	nationwide	ground
social	make	cause	customer	southwest
crisis	report	long	technical	issue
medium	Day	airport	delivery	problem
unforgettable	want	continue	fleet	stop
technology	travel	passenger	service	expect
hour	Time	experience	free	follow
suffer	people	line	break	love
hold	major	new	gift	lose
fix	Lot	pilot	buy	suck
week	book	deal	aircraft	work
	baltimore	year	bad	
	union	thank	way	
	leave	attendant	bag	
	story	impact	card	
	record	cancelation	good	
	receive	big	investigate	
	Sale	massive	departure	
	second	uniform	create	

(*continued on next page*)

Table 7 (*continued*)

Topic 1: "Flight Cancel" $N = 538$ (30.38%)	Topic 2: "Travel Plan" $N = 257$ (14.51%)	Topic 3: "Flight Delay Experience" $N = 350$ (19.76%)	Topic 4: "Customer Service" $N = 242$ (13.66%)	Topic5 "Emotional Experience" $N = 384$ (21.68%)
	Land	wide	company	
	Daily	beach	share	
	profit	photo	traveler	
	great	early	news	
	Site		stock	
	Right		remain	
	Fare		stick	
	Late		win	
	carrier		use	
			offer	

(4). Top words from the LDA model based on the probability of word occurrence ($T = 5$): Alaska Airlines

Topic 1: "Alaska Airline Brand" $N = 267$ (54.60%)	Topic 2: "Flight News" $N = 52$ (10.63%)	Topic 3: "Offer and deal $N = 59$ (12.06%)"	Topic 4: "Flight Experience" $N = 50$ (10.22%)	Topic5: "Non-stop Service"$N = 61$ (12.47%)
say	Fly	offer	flight	airline
brand	New	man	stick	stop
year	tentative	divert	wheel	announce
	post	try	plane	nonstop
		sleep	today	
		kick		
		free		
		girl		
		old		
		allegedly		
		family		

(5) Top words from the LDA model based on the probability of word occurrence ($T = 5$): Hawaiian Airlines

Topic 1: "Hawaiian Airlines Brand" $N = 165$ (61.80%)	Topic 2: "Seat and Class" $N = 14$ (52.43%)	Topic 3: "Kids related service" $N = 28$ (10.49%)	Topic 4: "Flight Experience" $N = 52$ (19.48%)	Topic5: "Service Upgrade" $N = 8$ (3.00%)
Airline	seat	fly	emergency	upgrade
Hawaiian	class	kid	make	auction
		economy	flight	
			landing	
			land	
			close	
			puncture	

ducted a variety of content analyses within the Twitter environment. Furthermore, to explore the effects of Twitter buzz on user engagement, we conducted the supervised learning procedure of Poisson regression as well as OLS regression (Table 8). By considering the unique characteristics of the Twitter dataset, we focused on the Poisson regression to test our research question.

In general, the gaps between the tweet generated date and the retweeted date (Twitter_RT Gap) showed a positive association with both the number of retweets and the number of likes. The number of hashtags showed a negative relationship with the dependent variables except for Alaska Airlines. Word count was a positive predictor of the number of retweets and the number of likes for United Airlines and Delta Airlines but was negatively associated for Alaska Airlines. In addition, the results were mixed for Southwest Airlines and Alaskan Airlines.

Interestingly, the results regarding the Twitter buzz sentiment were also mixed. For big airline companies, such as United Airlines and Delta Airlines, sentiments (i.e., compound score for each tweet) positively predicted both the number of tweets and the number of likes. However, for small airline companies, including Southwest Airlines, Alaska Airlines, and Hawaiian Airlines, the compound score was negatively associated with the number of retweets while either being positively related or non-significant with the number of likes.

4.4.1. Predicting the number of retweets

We display the Poisson regression results, including all independent and dependent variables, in Tables 8 and 9. Specifically, for United Airlines, the number of retweets increased by approximately 1.3% over time. The number of retweets increased by 54.7% when the sentiment of tweets was positive toward United Airlines. The number of retweets also increased by 2% when users used one more word. Compared to the baseline topic of brand-related issue, when Twitter users discussed United Airlines' special service, the number of retweets decreased by 26.1%. When Twitter buzz was related to a delayed complaint, the number of retweets increased by 35.7%. When Twitter users discussed United Airlines' flight attendant, the number of retweets increased by 14.3%. The pseudo R^2 was 10.8% (log likelihood: −17,267, Akaike information criterion: 34,552).

For Delta Airlines, the number of retweets increased by approximately 0.9% over time. The number of retweets increased by 9.3% when the sentiment of tweets was positive toward Delta Airlines and by 4.8% when one additional word was used. Compared to the baseline topic of service, when Twitter users discussed Delta Airlines' delayed complaints, the number of retweets increased by 66.1%. When Twitter users discussed Delta Airlines' bad flight experience, the number of likes decreased by 35.9%, whereas regarding the ticketing experience, the number of likes decreased by 116.5%. Finally, Twitter buzz regarding the passenger experience generated 41.1% more likes (pseudo R^2 = 10.95%, log likelihood = −5213, Akaike information criterion = 10,444).

Twitter buzz regarding Southwest Airlines indicated that the number of retweets increased by approximately 2.7% over time. The number of retweets decreased by 47.9% when the sentiment of tweets was positive toward Southwest Airlines. The number of retweets decreased by 2% when another hashtag was used and by 1.7% when an additional word was used. In terms of topics, compared to the baseline topic of flight cancelation, when Twitter users discussed Southwest Airlines related

Table 8
(1) The results of OLS and Poisson regression. Dependent variable: The number of retweets.

	United Airlines		Delta Airlines		Southwest Airlines		Alaska Airlines		Hawaiian Airlines	
	OLS	Poisson	OLS	Poisson	OLS	Poisson	OLS	Poisson	OLS	Poisson
Constant	−28.233	1.099***	−86.577	.853***	7.991**	1.615**	12.043	3.508***	−39.486*	1.669***
	(20.170)	(0.042)	(49.45)	(0.060)	(3.933)	(0.035)	(7.961)	(0.065)	(20.619)	(0.111)
Tweet_RT gap	.461***	.013***	.098 (0.099)	.009***	.284***	.027***	.023 (0.025)	.015***	.286***	.009***
	(0.069)	(0.0002)		(0.0003)	(0.020)	(0.0003)		(0.0004)	(0.108)	(0.001)
Compound	27.583	.547***	105.734**	.093**	4.125	−.479***	−3.891	−.761***	−6.081	−.279**
	(16.764)	(0.031)	(44.119)	(0.044)	(3.944)	(0.029)	(7.495)	(,071)	(26.098)	(0.121)
Hashtag Count	−.223	0.001	−1.900	−.004	−.263	−.020***	.359 (0.257)	.039***	−.347	.006 (0.004)
	(0.685)	(0.001)	(2.143)	(0.002)	(0.165)	(0.002)		(0.001)	(0.802)	
Word Count	1.466	. 020***	6.593**	.048***	−.187	−.017***	−.740	−.153***	3.843**	−.033***
	(1.372)	(002)	(3.335)	(0.003)	(0.323)	(0.002)		(0.005)	(1.754)	(0.008)
Topic 1	−15.435*	Baseline	45.664*	Baseline	4.650**	Baseline	3.292	Baseline	3.157	Baseline
	(9.173)		(25.794)		(2.358)		(3.602)		(11.629)	
Topic 2	−12.455	−.261***	−25.134	.661***	−.068	.885***	2.396	−.486***	2.847	−.580***
	(14.712)	(0.042)	(25.794)	(0.049)	(2.764)	(0.029)	(6.684)	(0.085)	(14.953)	(0.194)
Topic 3	25.381**	−.042	−77.373	−.359***	−2.209	.432***	1.483	.615***	−21.255	.198**
	(12.861)	(0.037)	(52.782)	(0.065)	(3.101)	(0.029)	(6.748)	(0.085)	(23.434)	(0.098)
Topic 4	−4.126	.357***	−10.370	−1.165***	6.999**	−.371***	1.418	−.296***	−12.313	.536***
	(13.244)	(0.026)	(51.802)	(0.111)	(3.443)	(0.041)	(6.379)	(0.085)	(19.494)	(0.078)
Topic 5	−21.607	.143***	−19.364	.411***	−1.381	.079**	1.414	−.108	−11.921	−.916***
	(17.813)	(0.031)	(41.461)	(0.054)	(2.957)	(0.031)	(8.570)	(0.067)	(15.793)	(0.292)
Observations	1591	588	1736	489	267					
R^2	.048	–	.024	–	.117	–	.044	–	.071	–
Pseudo R^2	–	.108	–	.109	–	.140	–	.175	–	.068
Log Likelihood	–	−17,267	–	−5213	–	−17,022	–	−5745	–	−1098
Akaike Information Criterion	–	34,552	–	10,444	–	34,062	–	11,508	–	2214

(2). The results of OLS and Poisson regression. Dependent variable: The number of Likes

	United Airlines		Delta Airlines		Southwest Airlines		Alaska Airlines		Hawaiian Airlines	
	OLS	Poisson	OLS	Poisson	OLS	Poisson	OLS	Poisson	OLS	Poisson
Constant	−19.007	.829***	−134.755	1.041***	7.706	1.900***	.198 (2.567)	1.851***	−28.81**	.902***
	(13.327)	(0.042)	(71.317)	(0.016)	(8.729)	(0.034)		(0.089)	(14.383)	(0.106)
Tweet_RT_ gap	.308***	. 015***	.100 (0.143)	.008***	.639***	.021***	−.016**	.012***	.209**	.003***
	(0.046)	(0.0002)		(0.0003)	(0.045)	(0.0003)	(0.008)	(0.001)	(0.075)	(0.001)
Compound	17.247	.475***	161.741**	.682***	16.107*	.437***	.584 (2.416)	.125 (0.083)	7.448	1.968***
	(11.078)	(0.028)	(63.623)	(0.042)	(8.753)	(0.028)			(18.205)	(0.105)
Hashtag Count	−.310	−.041***	−3.259	−.034***	−.54	−.035***	.131 (0.083)	.015***	−.337	−.027***
	(0.452)	(0.002)	(3.090)	(0.003)	(0.365)	(0.002)		(0.002)	(0.559)	(0.005)
Word Count	1.138	. 044***	10.363**	.051***	.024 (0.717)	.004*	−.032	−.057***	3.014**	.031***
	(0.906)	(0.002)	(4.809)	(0.003)		(0.002)	(0.186)	(0.006)	(1.224)	(0.007)
Topic 1	−9.276	Baseline	72.706*	Baseline	12.175**	Baseline	1.169	Baseline	1.089	Baseline
	(6.062)		(37.198)		(5.233)		(1.161)		(8.112)	
Topic 2	−8.071	−.833***	−38.173	−.620***	−2.877	.225***	.005 (2.155)	.174**	−.045	−1.383***
	(9.722)	(0.049)	(66.368)	(0.070)	(6.135)	(0.028)		(0.080)	(10.431)	(0.292)
Topic 3	16.059*	. 271***	−118.752	.197***	−4.586	−.199***	1.201	.453***	−12.007	−.516***
	(8.499)	(,034)	(76.117)	(0.049)	(6.88)	(0.028)	(2.176)	(0.071)	(16.347)	(0.160)
Topic 4	−3.305	.180***	−23.627	−1.895***	4.849	−.997***	−1.08	−.099	−5.850	.988***
	(8.752)	(0.028)	(74.704)	(0.118)	(7.641)	(0.042)	(2.057)	(0.095)	(13.598)	(0.076)
Topic 5	−14.413	. 627***	−26.901	.687***	−1.856	−.418***	−4.574*	.381***	−11.589	−1.631***
	(11.771)	(0.027)	(59.791)	(0.046)	(6.562)	(0.030)	(2.763)	(0.070)	(11.016)	(0.410)
Observations	1591	588	1771	489	267					
R^2	.051	–	.029	–	.116	–	.419	–	.042	–
Pseudo R^2	–	.141	–	.103	–	.854	–	.047	–	.162
Log Likelihood	–	−23,172	–	−7745	–	−25,145	–	−3.216	–	−1940
Akaike Information Criterion	–	46,363	–	15,508	–	50,308	–	6451	–	3898

to travel plans, the number of likes increased by 88.5%. Meanwhile, when flight delay experiences were discussed, the number of retweets increased by 43.2%, and when generating Twitter buzz regarding customer service, the number of retweets decreased by 37.1%. Finally, discussions regarding emotional experience generated 7.9% more retweets (pseudo R^2 = 14.03%, log likelihood = −17,022, Akaike information criterion = 34,062).

In terms of tweets related to Alaska Airlines, the number of retweets increased by approximately 1.5% over time, and the number of retweets decreased by 76% when the sentiment of tweets was negative toward Alaska Airlines. When tweets related to Alaska Airlines used more hashtags, the number of likes increased by 3.9%; however, the number of retweets decreased by 15.3% when one additional word was used. Compared to the baseline topic of Alaska Airlines' brand, when Alaska Airlines-related tweets discussed flight news, the number of retweets decreased by 48.6%. When offer and deal-related tweets were generated, the number of retweets increased by 61.5%. In addition, when Twitter buzz was related to the flight experience increased, the number of likes decreased by 29.6% (pseudo R^2 = 17.58%, log likelihood = −5745, Akaike information criterion = 11,508).

Table 9
Summary of Poisson regression results.

	United Airlines		Delta Airlines		Southwest Airlines		Alaska Airlines		Hawaiian Airlines	
	Retweet	Like	Retweet	Like	Retweet	Like	Retweet	Like	Retweet	Like
Tweet_RT gap	+	+	+	+	+	+	+	+	+	+
Compound	+	+	+	+	−	+	−	NS	−	+
Hashtag Count	NS	−	NS	−	−	−	+	+	NS	−
Word Count	+	+	+	+	−	+	−	−	−	+
Topic 1 (Reference)	"Business Related Issue"		"Service"		"Flight Cancelation"		"Alaska Airline Brand"		"Hawaiian Airlines Brand"	
Topic 2	"Special Service"		"Delay Complain"		"Travel Plan"		"Flight News"		"Seat and Class"	
	−	−	+	−	+	+	−	+	−	−
Topic 3	"Airport Service"		"Bad Flight Experience"		"Flight Delay Experience"		"Offer and Deal"		"Kids Related Service"	
	NS	+	−	+	+	−	+	+	+	−
Topic 4	"Delay Complain"		"Ticketing Experience"		"Customer Service"		"Flight Experience"		"Delay Complain"	
	+	+	−	−	−	−	−	NS	+	+
Topic 5	"Flight Attendant"		"Passenger Experience"		"Emotional Experience"		"Non-stop Service"		"Service Upgrade"	
	+	+	+	+	+	−	N.S	+	−	−

Note: + sign represents a positive relationship and **–** sign represents a negative relationship.

Regarding the Twitter buzz related to Hawaiian Airlines, the number of retweets increased by approximately 0.9% over time but decreased by 27.9% when the sentiment of tweets was positive toward Hawaiian Airlines. The number of retweets decreased by 3.3% when an additional word was added. Compared to the baseline topic of the Hawaiian Airlines brand, Hawaiian Airlines-related tweets discussing topics related to seat and class decreased the number of retweets by 58%. When users discussed kids-related service, the number of likes increased by 19.8%; when discussing delayed complaints, the number of retweets increased by 53.6%. However, when users generated buzz regarding service upgrades, the number of likes decreased by 91.6% (pseudo R^2 = 6.83%, log likelihood = −1098, Akaike information criterion = 2214).

4.4.2. Predicting the number of likes

We also fitted the Poisson regression model to estimate the effects of independent variables to predict the number of likes. For United Airlines, the number of likes increased by approximately 1.5% over time. When a tweet's or retweet's sentiment toward United Airlines was positive, the number of likes increased by 47.5%. As people or United Airlines used more hashtags, the number of likes decreased by 4.1%, whereas the number of likes increased by 4.4% when an additional word was used. Compared to the baseline topic of a business-related issue, when users discussed a special service, the number of likes decreased by 83.3%. When Twitter buzz was related to the airport service, the number of likes increased by 27.1%. When the topic related to delayed complaints, the number of likes increased by 18%. Topics related to flight attendants also increased the number of likes by 62.7% (pseudo R^2 = 14.1%, log likelihood = −23,172, Akaike information criterion = 46,363).

Regarding Delta Airlines' Twitter buzz, the number of likes increased by approximately 0.8% over time. When a tweet's or retweet's sentiment toward Delta Airlines was positive, the number of likes increased by 68.2%; when the number of hashtags increased, the number of likes decreased by 3.4%. The number of likes increased by 5.1% when an additional word was used. Compared to the baseline topic of service, when Twitter users and Delta Airlines discussed delayed complaints, the number of likes decreased by 62.0%. When Twitter buzz was mainly related to a bad flight experience, the number of likes increased by 45.3%, whereas ticketing experience buzz decreased the number of likes by 189.5%. In addition, tweets about Delta Airlines' passenger experience increased the number of likes by 68.7% (pseudo R^2 = 10.27%, log likelihood = −7745, Akaike information criterion = 15,508).

In terms of Southwest Airlines-related tweets, the number of likes increased by approximately 2.1% over time. When a tweet's or retweet's sentiment toward Southwest Airlines was positive, the number of likes increased by 43.7%. A greater use of hashtags decreased the number of likes by 3.5%. Compared to the baseline topic of flight cancelation, when Twitter users and Southwest Airlines discussed the topic travel plan, the number of likes increased by 22.5%. As Twitter buzz related to the flight delay experience, the number of likes decreased by 19.9%, and the customer service topic was associated with a 99.7% decrease in the number of likes. When Twitter users and Southwest Airlines discussed the topic of emotional experience, the number of likes decreased by 41.8% (pseudo R^2 = 8.54%, log likelihood = −25,145, Akaike information criterion = 50,308).

Regarding Alaska Airlines-related Twitter buzz, the number of likes increased by 1.2% over time. When users employed additional hashtags, the number of likes increased by 1.5%, whereas the number of likes decreased by 5.7% when an additional word was used. Interestingly, the compound score (i.e., the level of positivity of each tweet) was not associated with the number of likes. Compared with the baseline topic of Alaska Airlines brand, when Twitter users and Alaska Airlines discussed flight news, the number of likes increased by 17.4%; when discussing offers and deals, the number of likes increased by 45.3%. The Twitter buzz associated with non-stop service increased the number of likes by 38.1% (pseudo R^2 = 4.72%, log likelihood = −3.216, Akaike information criterion = 6451).

Finally, regarding the results related to Hawaiian Airlines, the number of likes increased by approximately 0.3% over time. When the sentiment toward Hawaiian Airlines was positive, the number of likes increased by 196.8%. When users used more hashtags, the number of likes decreased by 2.7%; however, the number of likes increased by 3.1% when an additional word was used. Compared to the baseline topic of Hawaiian Airlines' brand, when Twitter users and Hawaiian Airlines discussed seats and class, the number of likes decreased by 138.3%. The Twitter buzz related to the kids-related service decreased the number of likes by 51.6%. When Twitter users and Hawaiian Airlines discussed delayed complaints, the number of likes increased by 98.8%, whereas

Table 10
Comparison of topic modeling results and ACSI survey results.

Topics	LDA topic modeling (This research)	ACSI 2022	
	Included	Included	Score
Delay complains	√	√	81
Ticketing experience	√	√	81
Flight cancels	√	√	81
Flight attendant	√	√	80
Seat and class	√	√	72
Travel plan	√	×	×
Special service	√	×	×
Airport service	√	×	×
Bad flight experience	√	×	×
Passenger experience	√	×	×
Emotional experience	√	×	×
Offer and deal	√	×	×
Non-stop service	√	×	×
Airline company brand	√	×	×
Kids related service	√	×	×
Service upgrade	√	×	×

Note: √ = included, × = not included.

discussions of service upgrade were associated with a 163.1% increase in the number of likes (pseudo R^2 = 16.2%, log likelihood = −1940, Akaike information criterion = 3898).

5. Discussion

5.1. Summary of findings

5.1.1. RQ1: what was the prominent topic of the Twitter buzz, and how did the topics differ across the airline companies?

From a study of the Twitter buzz related to five different airline companies, our findings clearly supported the existence of differences across prominent topics from different companies. For relatively big companies, such as United Airlines and Delta Airlines, Twitter buzz was mainly related to the service experience, such as special services for traveling with a dog, airport services, ticketing experience, and flight experience. For smaller companies, such as Alaska Airlines and Hawaiian Airlines, consumers mentioned more unique aspects of the attributes that could be utilized as a point of difference, such as seats, kids-related services, upgraded seats, and offers and deals (Table 9). We compare our findings with the ACSI survey results in Table 10.

5.1.2. RQ2: what was the valence of the Twitter buzz, and how did the sentiment differ across the airline companies?

By adopting a supervised learning process of VADER for sentiment analysis, our analysis results found a discrepancy between the results from the structured survey (American Customer Satisfaction Index: ACSI, American Customer Satisfaction Index, 2019) and our unstructured data analysis. For example, based on our findings, Twitter buzz from United Airlines was classified as the most positive sentiment (0.37, ACSI: 77), followed by Alaska Airlines (0.13, ACSI: 75), Delta Airlines (0.06, ACSI: 77), Southwest Airlines (−.03, ACSI: 77), and Hawaiian Airlines (−.03, ACSI: N.A). This discrepancy might have occurred due to the types of situations that prompted customers to express an opinion using social media. In general, customers express their opinion on social media in response to extreme conditions. For example, when customers experience extreme disappointment at the airport, they might open Twitter and express their anger instantly (Ghezzi et al., 2016). Therefore, by its nature, sentiments and opinions expressed through social media are more extreme and dispersed (He & Bond, 2015; Rozenkrants et al., 2017) (Table 11).

5.1.3. RQ3: what was the effect of Twitter buzz on consumer engagement, and how did the effects of Twitter buzz differ across the airline companies?

Most interestingly, we estimated the effects of our independent variables on the user engagement level by employing the more rigorous estimation procedures of Poisson regression. First, our results of estimation showed that the gaps between the tweet generated date and the retweeted date were positively associated with both the number of retweets and number of likes.

Second, generally speaking, the number of hashtags showed a negative relationship with both the number of retweets and the number of likes, except for Alaska Airlines. This finding suggests that utilizing many hashtags generally produced a backfire effect and was not helpful in boosting users' engagement level.

Third, we found mixed effects regarding the number of words used within the tweet. For relatively big companies. such as United Airlines and Delta Airlines, we captured positive associations between the word count and the dependent variables. In contrast, for small companies, such as Southwest Airlines and Hawaiian Airlines, increasing the word count negatively predicted the number of retweets but was positively associated with the number of likes. This suggests that small airline companies need to separate social media message strategies based on the goal of the social media marketing—namely, whether the goal is increasing favorable attitude toward the message (i.e., the number of likes) or maximizing the virality of the message (e.g., the number of retweets).

Fourth, in terms of the effects of sentiments on users' engagement levels, we also found mixed results based on firm size. For big companies (e.g., United Airlines and Delta Airlines), the compound score of the tweets (i.e., sentiment of tweets) was positively associated with both dependent variables. However, for relatively small airline companies, the compound score had a negative association with the number of retweets, whereas the compound score positively predicted the number of likes. This finding suggests that users discussing topics related to Southwest Airlines, Alaska Airlines, and Hawaiian Airlines generated more likes when the theme of the tweet was positive than when they spread messages containing a negative sentiment (e.g., complaints).

Finally, this study found different effects of company-specific themes for predicting user engagement. For example, for the United Airlines, topics related to delayed complaints and flight attendants generated more retweets and likes, whereas buzz related to ticketing experience for Delta Airlines had a negative association with user engagement (Tables 3–5).

5.2. Theoretical contributions

5.2.1. Antecedents of eWOM effectiveness and understanding user engagement

Our novel conceptualization of the antecedents of user engagement, based on the eWOM perspective, revealed the empirical relationship between the proposed features and user engagement. It added to the extant research that examines the effects of Twitter message components on user engagement, mainly addressing the emotional aspect of message (Previte et al., 2019) or source characteristics (Hautz et al., 2014). By adding several linguistic and message design components—such as the number of words, the gaps between tweet generated date and the retweeted date, and the number of hashtag(*s*)-we demonstrated the different effects of such factors on user engagement level.

This research also added knowledge regarding the effects of eWOM valence. Previous studies revealed the effects of the eWOM valence on attitude formation and its virality (Berger & Milkman, 2012). Rooted in the negativity bias perspective (Rozin & Royzman, 2001), previous eWOM and impression formation literature found stronger effects of negative eWOM messages when compared to positive eWOM messages, as negative information about a person or a brand carries more weight than positive ones. In addition, Berger and Milkman (2012) found that positive content became more viral by analyzing news article sharing

Table 11
Comparison of sentiment level form this study and satisfaction score from ACSI.

This research			ACSI 2022		
Rank	Airline company	Sentiment (compound) score	Rank	Airline company	Satisfaction score
1.	United Airlines	.37	1	Delta Airlines	77
2	Alaska Airlines	.13	1	Southwest Airlines	77
3	Delta Airlines	.06	1	United Airlines	77
4	Southwest Airlines	−.03	4	Alaska Airlines	75
4	Hawaiian Airlines	−.03			

behavior. However, our findings regarding consumer sentiment contradicted these findings by showing the moderating effects of company size. For large airline companies, such as United Airlines and Delta Airlines, positive sentiment predicted an increase of user engagement, whereas for relatively small airline companies, the results revealed the mixed effects of sentiment. Specifically, the compound score was negatively associated with the number of retweets, whereas it positively predicted the number of likes.

5.2.2. Causal inspection of generated features on the dependent variables

Although previous literature in the information system and marketing fields has adopted the computerized text analysis method (e.g., Misopoulos et al., 2014; Takahashi et al., 2015), most previous studies focused solely on the meaning extraction procedure of unstructured data. We further added the causal inspection process between generated topics and user engagement level.

5.3. Managerial implications and strategic defaults

Understanding the dynamics of how user engagement is formed is one of the most important processes for managers to understand to build successful marketing strategies for airline companies. Our findings apply to both marketing/information system practitioners who want to build a successful communication strategies and marketers who work in customer relationship management (CRM). We discuss implications for both.

5.3.1. For communication strategy

Our research provides a guide or a template for social media marketing/information system development Based on our findings, advertisers must strategically determine the optimal combination of message features. To design a social media message for the large airline companies, managers should emphasize positively framed themes to get a more favorable reaction toward the brand and boost the virality of the message.

In marketing large airline companies, managers should use more words to generate tweets. For example, for Delta Airlines, managers wanting to emphasize the positive passenger experience should add more descriptions or details to generate more user engagement. However, for small airline companies, using simple and concise messaging is likely to generate more positive reactions.

5.3.2. For social CRM strategy

Contrary to the structured survey results, the Twitter buzz for Southwest Airlines and Hawaiian Airlines was negative. Therefore, they should do more active CRM and communication campaigns in the social media environment.

For many airline companies, discussion regarding the “delayed complaint” was generally going more viral, and these conversations were getting more attention. Therefore, airline companies should focus on this issue to get a higher level of customer satisfaction.

We captured the unique themes for a given airline company. For instance, for United Airlines, users frequently mentioned the “flight attendant,” and users that mentioned Hawaiian Airlines discussed the unique topic of “kids related service.” Therefore, marketers can utilize the “kids related to service” as a unique point for Hawaiian Airlines to position its brand and differentiate from its competitors.

5.4. Limitations and future research directions

Despite this study’s setting and methodology, which allowed us to accurately measure the effects of various message components of social media on user engagement, this work is not without limitations. Our results only showed the effects of text features on Twitter. Considering the effects of image on user engagement and attitude formation, additional research is needed to analyze the effects of image features and their interaction effects with text features.

Also, we hope that follow-up research will employ additional machine learning techniques for the natural language process, such as Word2vec (Mikolov et al., 2013; Park et al., 2019). Further research should study in greater depth how topics evolve overtime. We employed the unsupervised learning process of LDA, which focused on the cross-sectional aspect of data, although we collected the data within 95-day windows. Therefore, by adopting time series analysis approaches, such as the nonparametric mixture model (Dubey et al., 2013) or Chinese restaurant processes (Blei & Frazier, 2011), further examination could expand this research context to include changes in prominent topics overtime.

In addition, future studies should examine the utilization of different AI-related technologies. The current study utilized supervised and unsupervised learning processes, which are the core component of AI, but conceptually, the adoption of AI applications in a variety of contexts should be revisited (Almaiah et al., 2022a, 2022b).

6. Conclusions

In summary, our findings suggested that social media communication strategy should consider multiple dimensions including themes that are frequently discussed among users and the characteristics of the message. Since the importance of eWOM has significantly increased overtime, more resources and effort should be allocated to collect, analyze, and utilize unfiltered consumer opinion through social media platforms. Therefore, this study provided a detailed guide for mangers on how to utilize an unstructured data analysis procedure to adopt both supervised and unsupervised learning processes, yet much remains to be learned to realize the full potential of AI-related technologies in understanding consumer satisfaction.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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